QUESTIONS

Q1
Carbon-Aware Cloud Computing Management System
50 marks

Q2
Smart Renewable Energy and EV Charging System
50 marks

2/2 answered

Q1 Carbon-Aware Cloud Computing Management System

50 marks

A cloud service provider is modernizing its Carbon-Aware Cloud Management System to reduce the environmental impact of computing workloads. The system manages AI Workloads, Web Applications, and Data Analytics Workloads. Current issues include high energy consumption, inflexible energy-management strategies, difficulty supporting new workload categories, inefficient resource management, and inflexible carbon-emission calculation. Develop a C++ OOP application using inheritance, virtual base classes, abstract classes, constructors in derived classes, composition, pointers, the this pointer, pointers to derived classes, and runtime polymorphism. Design an abstract base class Workload with AIWorkload, WebWorkload, and AnalyticsWorkload. Create CriticalWorkload using multiple inheritance from AIWorkload and AnalyticsWorkload, use virtual base classes to resolve ambiguity, and include an EnergyMonitor member class. Implement pure virtual calculateCarbon(). Demonstrate Workload*, new, delete, this pointer, pointer to derived class, and a menu-driven program.

Your Code

```
#include <iostream>
using namespace std;

class EnergyMonitor {
public:
    void show() { cout << "Energy monitored\n"; }
};

class Workload {
protected:
    double energy, factor;
public:
```

Input

stdin input

Output

Q1
Carbon-Aware Cloud Computing Management System
50 marks

Q2
Smart Renewable Energy and EV Charging System
50 marks

2/2 answered

Q2 Smart Renewable Energy and EV Charging System

50 marks

A smart city is developing a Renewable Energy and EV Charging Management System integrating solar power, battery storage, and EV charging stations. Current issues include uneven energy distribution, difficulty managing energy sources, inflexible charging-cost calculation, difficulty supporting future renewable technologies, and inefficient dynamic handling. Develop a scalable C++ OOP solution using inheritance, multiple inheritance, virtual base classes, abstract classes, constructors and destructors, composition, pointers, this pointer, pointer to derived classes, and runtime polymorphism. Design abstract EnergySource with SolarEnergy, WindEnergy, and BatteryStorage. Create HybridChargingStation using multiple inheritance, use a virtual base class, and include a ChargingMonitor member class. Implement pure virtual calculateEnergyCost(). SolarEnergy uses solar energy � charging rate; WindEnergy uses wind energy � charging rate; BatteryStorage uses stored energy � discharge rate. Demonstrate EnergySource*, new, delete, pointer to derived class, this pointer, polymorphism, and a menu-driven program.

Your Code

```
#include <iostream>
using namespace std;

class ChargingMonitor {
public:
    void monitor() {
        cout << "Charging monitored\n";
    }
};

class EnergySource {
protected:
```

Input

```
stdin input
```

Output